Name: _____ Period: ____

Horizontal Projectiles

Honors Physics · Unit 2 · Bridge lab: from relative velocity to horizontal launches · 45 minutes

Work in groups of three. In Sections 2.5 and 2.6 a boat or plane had two velocities at once, and you added them as vectors. Today you look for the same idea in a falling ping pong ball and a marble rolling off a table. Predict before every test, and write the prediction down before anyone in your group tries it.

Equipment per group: a ping pong ball, a marble, a lab table, masking tape, two meter sticks and a textbook (the track), a third meter stick or tape measure, carbon paper and a sheet of white paper, and a phone that records slow-motion video.

Warm up · A kayak crossing

5 min

Mr. Jantz paddles a kayak at 2.0 m/s straight across a river 60 m wide. The current flows at 1.5 m/s downstream. He keeps the kayak pointed straight across the whole time.

1. How long does the crossing take? How far downstream does he land?

2. Find his velocity relative to the ground (size and direction).

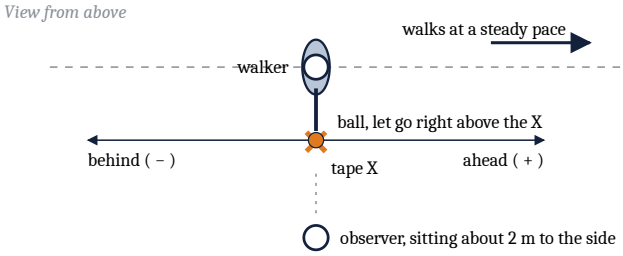
3. The current doubles to 3.0 m/s. What happens to the crossing time? To the landing spot? Explain in one sentence.

Connect to Sections 2.5 and 2.6. Two velocities that point in perpendicular directions each do their own job. Keep an eye out today for a motion that "sets the clock" and a motion that just rides along.

Part 1 · The walking drop

6 min

Put a tape X on the floor. The **walker** holds a ping pong ball at shoulder height, out to the side. The **observer** sits on the floor about 2 m to the side, facing the X, and marks the landing spot.



4. **Predict.** The walker walks past the X at a steady pace and simply opens their hand when the ball is right above the X. Where does the ball first hit the floor: on the X, behind it, or ahead of it? Why? (Measure to where the ball first touches the floor, not where it stops rolling. The observer marks that spot with tape.)

Test. The walker walks past at a steady pace and lets go above the X. Do 3 trials and record each landing spot (+ ahead, − behind).

Trial 1: _____ m Trial 2: _____ m Trial 3: _____ m

5. Where was the walker's hand when the ball landed: behind the ball, above it, or ahead of it? What does that tell you about the ball's forward motion while it fell?

The ball's path as the walker sees it

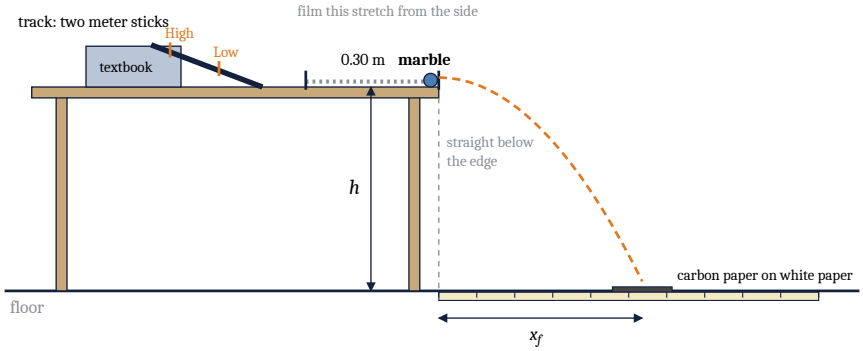
The path the observer sees (it is fast, so sketch your best reasoning)

6. Use the relative velocity idea from Section 2.5: the ball's velocity relative to the ground equals the ball's velocity relative to the walker plus the walker's velocity relative to the ground, $\vec{v}_{\text{ball, ground}} = \vec{v}_{\text{ball, walker}} + \vec{v}_{\text{walker, ground}}$. The walker moves at 1.2 m/s. Right after release, what is the ball's horizontal velocity relative to the walker? Relative to the ground?

Connect to Section 2.5. In the Relative Velocity Lab, the **raft** view showed the boat going straight across while the **shore** view showed it moving diagonally. The walker here is the raft, and the observer is standing on the shore.

Part 2 · Off the edge

20 min



Lay two meter sticks side by side, a marble-width apart, with one end propped on a textbook. The bottom end should sit about 0.40 m from the table edge. Tape off the last 0.30 m before the edge, and mark a **Low** and a **High** release spot on the track. Tape white paper on the floor where the marble lands, with carbon paper on top, face down.

7. Measure the height of the tabletop above the floor.

h = _____ m

NOTE · SLOW-MOTION TIMING Film the taped 0.30 m stretch from the side in slow motion. Play it back and read the time on your phone when the front of the marble crosses the start tape and when it reaches the edge. Subtract to get the time to roll 0.30 meters.

For each release spot, film one roll to find the launch speed v_{ix}, then do 3 landing trials from the same spot. Record the speed in the first table. In the second, x_f is the landing distance: from the spot directly under the table edge to the carbon mark.

Release	**Time to roll 0.30 meters (s)**	**Launch speed:** $v_{ix} = \dfrac{0.30\text{ m}}{\text{time}}$ **(m/s)**
Low		
High		

Release	**Trial 1** x_f **(m)**	**Trial 2** x_f **(m)**	**Trial 3** x_f **(m)**	**Average** x_f **(m)**	**Time in the air:** $t = \dfrac{\text{average } x_f}{v_{ix}}$ **(s)**
Low					
High					

8. Look at both of your data tables. What patterns do you see? Describe as many as you can.

9. Use Unit 1 free fall to predict how long a marble dropped from rest at your table edge takes to reach the floor. Fill in the y column, then solve for t in the space on the right.

Vertical (y)	Value
y_i	
y_f	
v_{iy}	
a_y	
t	

Solve for t:

10. How does your predicted fall time compare with the time in the air from your landing table? What does that tell you about the rolling marble's fall?

Think back. In the Horizontal Projectile Lab, a ball launched sideways and a ball dropped from the same height landed at the same time. MythBusters got the same result with two bullets: one fired level and one dropped by hand hit the ground together. Your marble data shows the same thing: launching sideways does not change the fall time. What the sideways speed changes is how far the object travels.

11. Build an equation for the landing distance.

a. Once the marble leaves the table, is anything pushing it forward or backward? (Think about what touches it.)

b. So while the marble is in the air, is its horizontal velocity speeding up, slowing down, or constant? What is a_x?

c. In Unit 1, an object moving at a constant velocity covers distance = velocity × time. The marble starts at the edge ($x_i = 0$). Write an equation for x_f using v_{ix} and t.

d. Check it: multiply your Low v_{ix} by your fall time from Question 9. How close is it to your average x_f for Low?

12. Your measured average x_f and your prediction $v_{ix}t$ probably do not match exactly. Name two things that could cause the difference. For each, say whether it would make the measured x_f smaller or larger than predicted.

Part 3 · Velocity vectors in flight

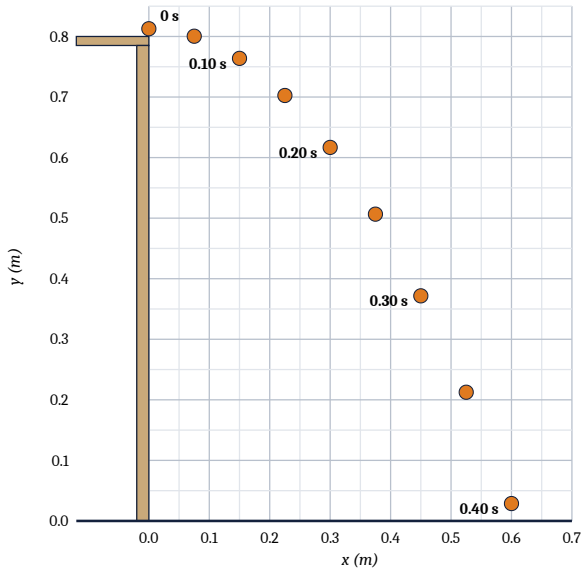
10 min

The picture shows a ball rolling off a 0.80 m table at 1.50 m/s, drawn every 0.050 s. Grid lines are 0.05 m apart.

13. Find the horizontal distance between neighboring dots. Is it the same each time? Use it to find v_x.

14. Look at the vertical drop between neighboring dots. Does it stay the same, grow, or shrink? Which Unit 1 motion does that look like?

15. Fill in the table. Use $v_{fy} = a_y t + v_{iy}$ with $v_{iy} = 0$ and $a_y = -9.8\ \text{m/s}^2$. Find each speed and angle the way you found resultants in Section 2.4.



t (s)	v_x (m/s)	v_y (m/s)	Speed (m/s)	Angle below the +x-axis
0	1.50	0	1.50	0°
0.10				
0.20				
0.30				
0.40				

16. On the figure, draw the velocity at the 0.30 s dot tip-to-tail: a horizontal arrow for v_x, then a vertical arrow for v_y from its tip, then the resultant. Which way does the resultant point compared with the path?

17. **Your marble.** Use your High v_{ix} and your fall time from Question 9. Find v_x and v_y just before the marble hits the floor, then its speed and its angle below the +x-axis.

Connect to Sections 2.3 and 2.4. Like the kayak, the ball's velocity has two independent components. The kayak's were both constant, so its path was straight; here v_y keeps growing, so the path curves.

Wrap up · Same idea, new direction

4 min

	Kayak crossing a river	Marble rolling off a table
What sets the time?	River width and the across velocity	
Does the sideways speed change the time?	No	
Sideways distance	drift $= v_{\text{current}}\, t$	
Shape of the path, and why	Straight: both velocities are constant	

18. **Exit check.** A marble rolls off your table at 3.0 m/s. Without a new test, predict its time in the air and its landing distance x_f. Explain using your data.